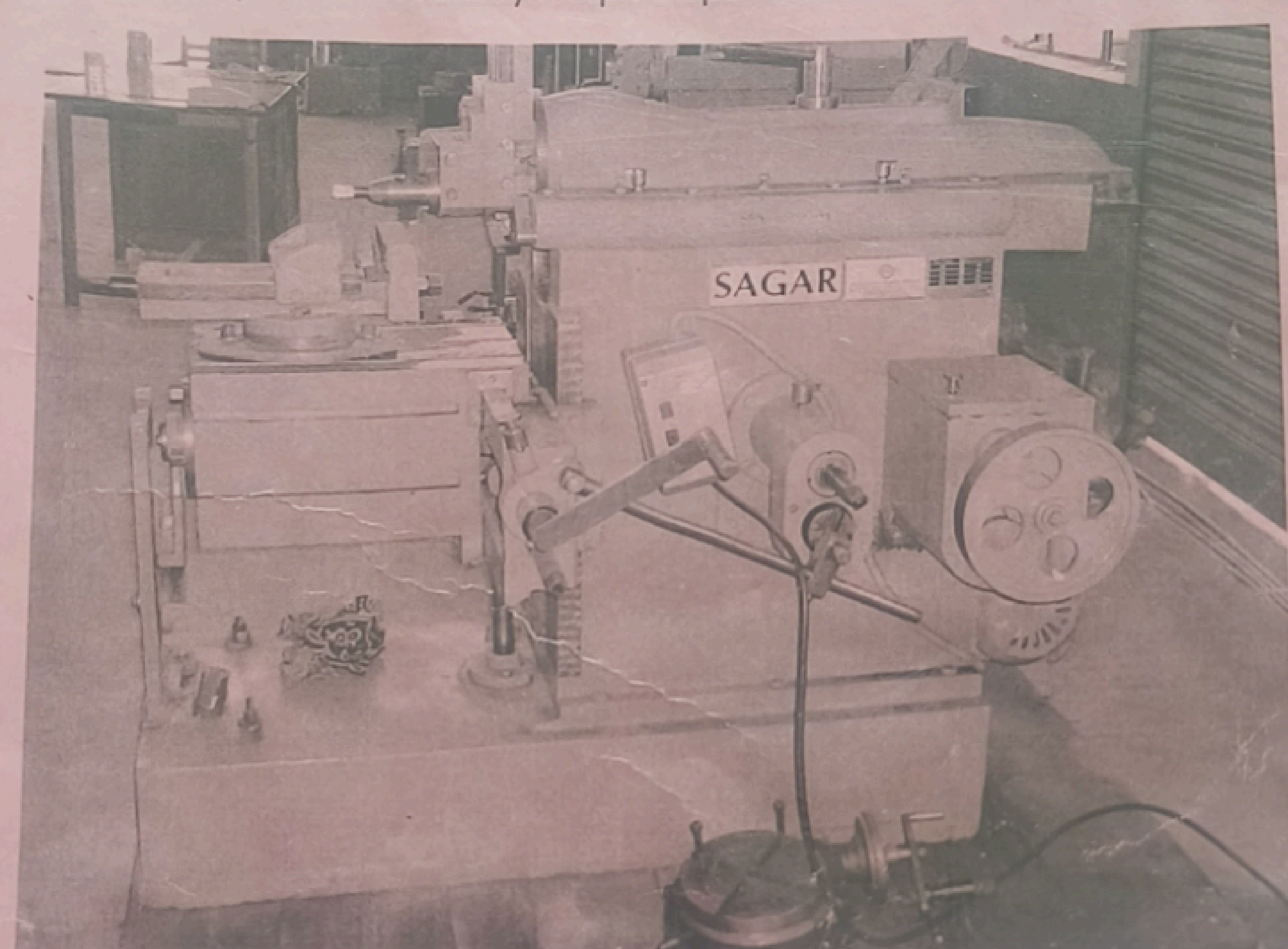
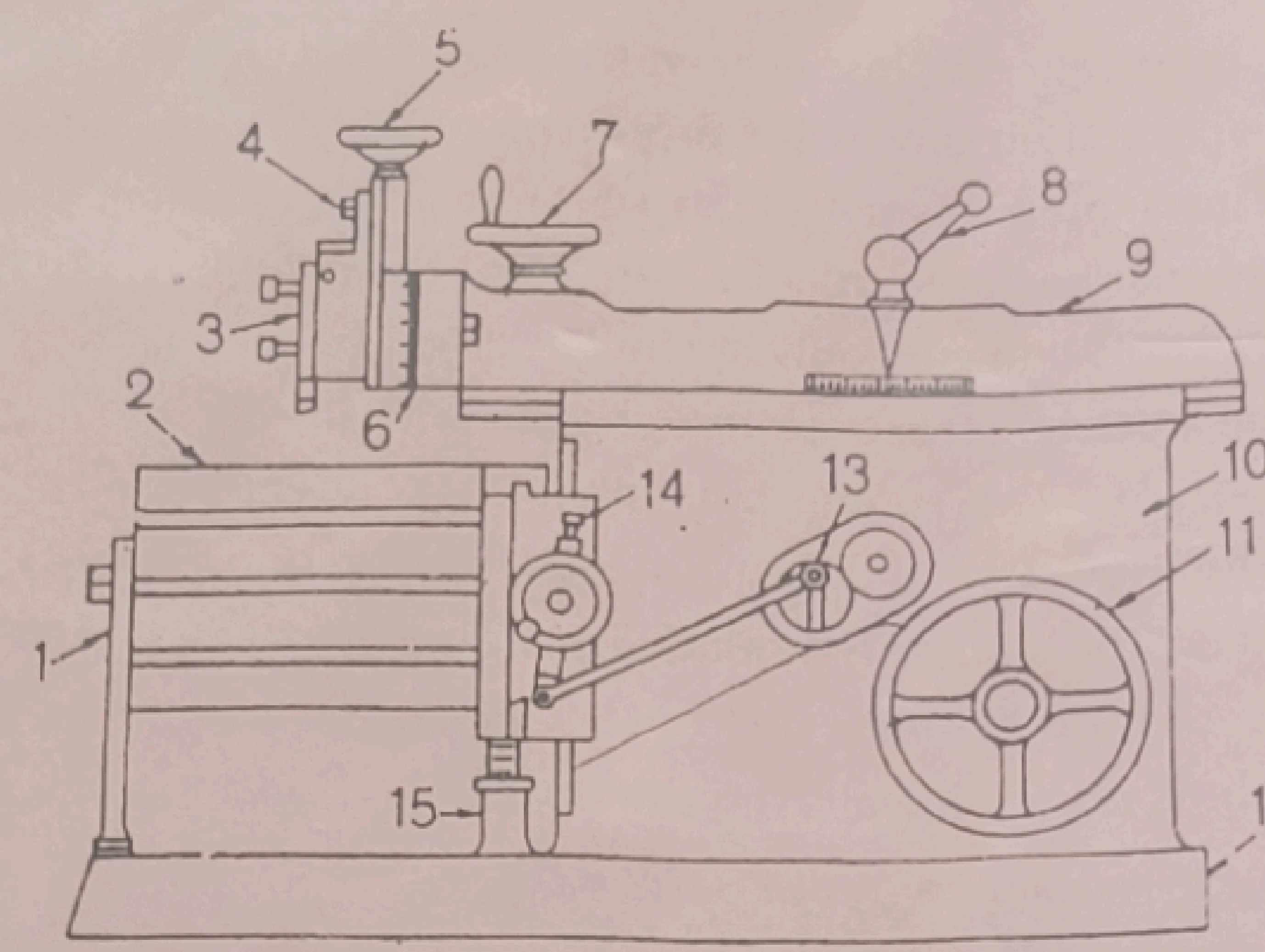
THE SHAPER MACHINE

The shaper is a relatively simple machine. It is used fairly often in the tool room or for machining one or two pieces for prototype work. The metal working shaper was developed by James Nasmyth in 1836. Shaper is a machine tool to produce flat surfaces. The flat surfaces may be horizontal, vertical or inclined. The shaper uses single point tools and cuts only in straight lines. The shaper handles relatively small work. The cutting stroke of the shaper is made by moving the tool bit attached to the ram. The shaper cut only in one direction, so that the return stroke is lost time. However, the return stroke is made at up to twice the speed of the cutting stroke. The shaper is a relatively simple machine. It is used fairly often in the tool room or for machining one or two pieces for prototype work. Tooling is simple, and shapers do not always require operator attention while cutting.



A horizontal shaper

COMPONENTS OF A SHAPER MACHINE:



1. Table support, 2. Table, 3. Clapper block, 4.Apron clamping bolt, 5. Downfeed hand wheel, 6. Swivel base degree graduations, 7. Position of stroke adjustment, 8. Ram block locking handle, 9. Ram, 10. Column, 11. Driving Pulley, 12. Base, 13. Feed disc, 14. Pawl mechanism, 15. Elevating screw

Ram: The ram slides back and forth in dovetail or square ways to transmit power to the cutter. The starting point and the length of the stroke can be adjusted.

Toolhead: The toolhead is fastened to the ram on a circular plate so that it can be rotated for making angular cuts. The toolhead can also be moved up or down by its hand crank for precise depth adjustments. Attached to the toolhead is the tool holding section. This has a tool post very similar to that used on the engine lathe. The block holding the tool post can be rotated a few degrees so that the cutter may be properly positioned in the cut.

Clapper Box: The clapper box is needed because the cutter drags over the work on the return stroke. The clapper box is hinged so that the cutting tool will not dig in. Often this clapper box is automatically raised by mechanical, air, or hydraulic action.

Adjustment of length of stroke:

The distance of the sliding block from the centre of the bull gear can be varied radially within the radial slide. If the sliding block is brought inwards i.e. towards the centre the length of stroke will be reduced, and vice versa.

Adjustment of the position of stroke:

The position of the ram relative to the work can also be adjusted. The rocker arm is connected to a screw on the ram. The position of this connection can be varied on this screw. So the ran can be positioned adjusting this by means of a lever and wheel provided on top of the ram.

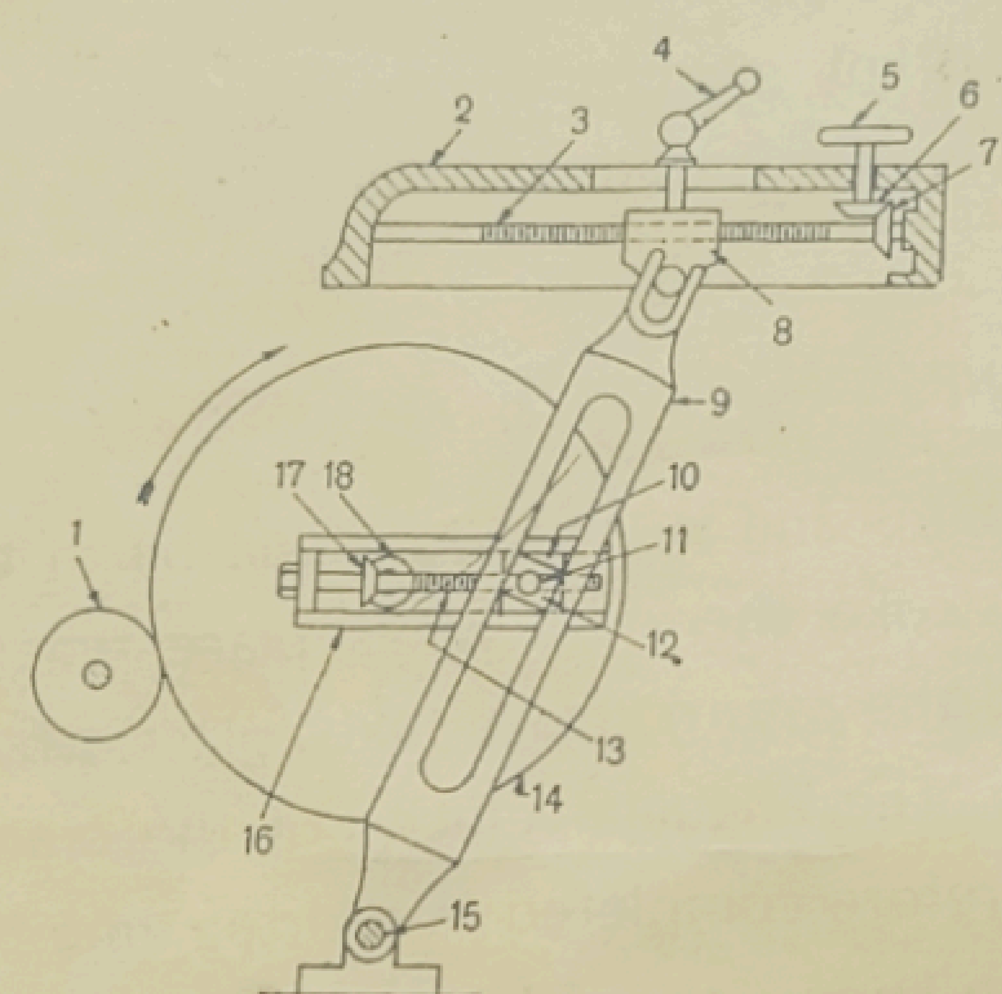
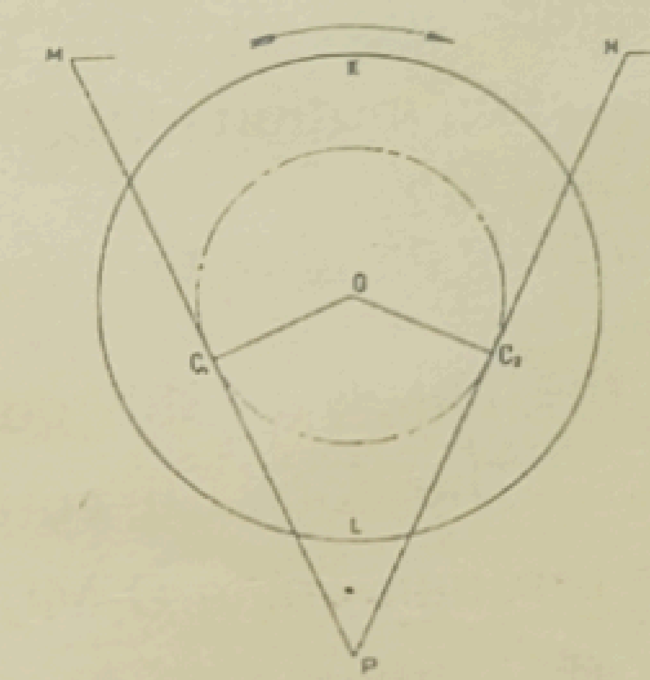


Fig. 7.3. Crank and slotted link mechanism
1. *Driving pinion*, 2. *Ram*, 3. *Screwed shaft*, 4. *Clamping lever*, 5. *Handwheel for position of stroke adjustment*, 6,7. *Bevel gears*, 8. *Ram block*, 9. *Slotted link or rocker arm*, 10. *Bull gear sliding block*, 11. *Crank pin*, 12. *Rocker arm sliding block*, 13. *Lead screw* 14. *Bull gear*, 15. *Rocker arm pivot*, 16. *Bull gear slide*, 17,18. *Bevel gears.*



$$\frac{\text{Cutting time}}{\text{Return time}} = \frac{\widehat{C_1KC_2}}{\widehat{C_2LC_1}}$$

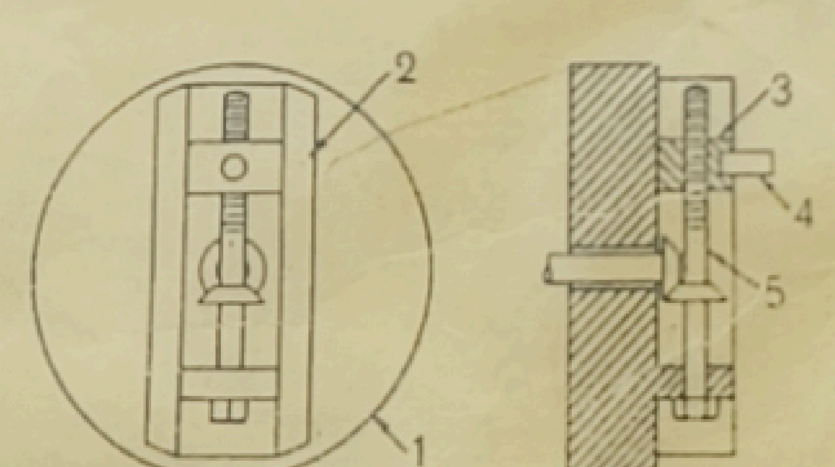


Fig. 7.5. Buli gear sliding block mounting arrangement
1. *Bullgear*, 2. *Bullgear slide*, 3. *Bull gear sliding block*, 4. *Crank pin*, 5. *Lead screw.*

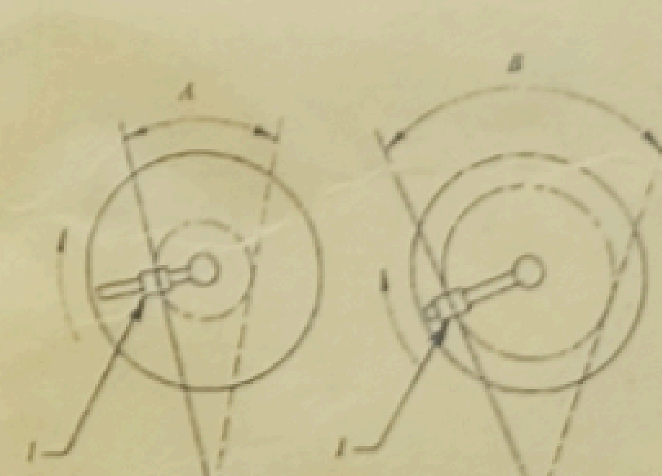


Fig. 7.6. Stroke length adjustment
1. *Position of crankpin*, *A. Short stroke length*, *B. Long stroke length*

Crank and Slotted Link Mechanism

The bull gear is driven by a pinion which is connected to the motor shaft through a gear box with four, eight or more speeds available. The speed of the bull gear may be changed by different combination of gearing or by simply shifting the belt on step cone pulley. Bolted to the centre of the bull gear is a radial slide (not shown in figure). This slide carries a sliding block, into which the crank pin is fitted. The crank pin connects the slotted link and the sliding block. The slotted link which is known as rocker arm is pivoted at the bottom end attached to the frame of the column. The upper end of the rocker arm is connected to the ram. As the bull gear rotates causing the crank pin to rotate, the sliding block fastened to the crank pin will rotate on the crank pin circle, and at the same time move up and down the slot in the slotted link giving it a rocking movement which is communicated to the ram. Thus the rotary motion of the bull gear is converted to reciprocating motion of the ram.

The principle of quick return motion:

When the link is in the position AB and AC, the ram will be in the extreme positions. The forward cutting stroke therefore takes place when the crank rotates through angle D_1MD_2 and the return stroke takes place when the crank rotates through the angle D_1LD_2. The angular velocity of the crank pin being constant the return stroke is therefore completed in shorter time than forward stroke. Therefore this mechanism is known as quick return mechanism.

Therefore, $\dfrac{\text{Cutting time}}{\text{Return time}}$

Cutting time to return time ratio varies within 2:1 and the practical limit is 3:2.

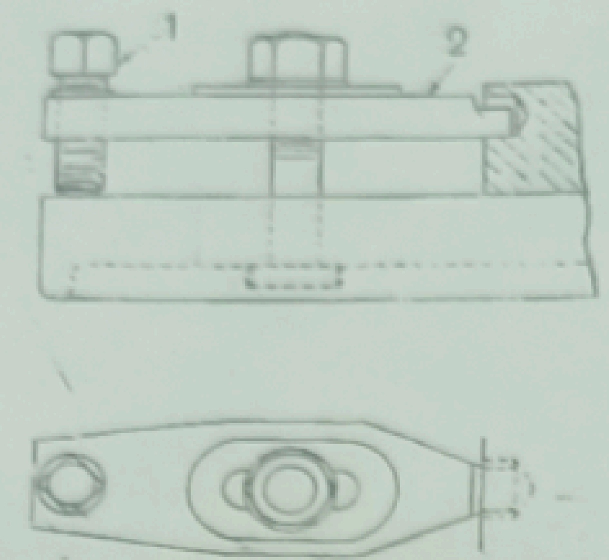


Fig. 5.10 Use of adjustable step clamp
1. *Adjusting screw*, 2. *Clamp*.

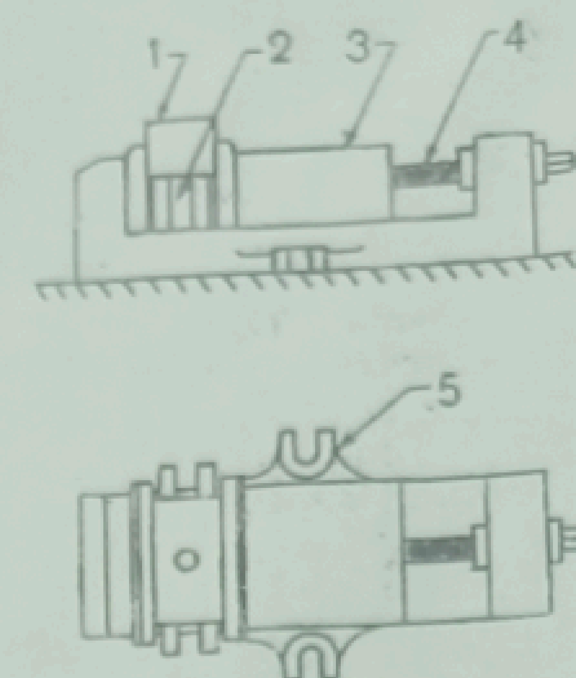


Fig. 5.11 Plain vise
1. *Work*, 2. *Parallels*, 3. *Movable unit* 4. *Screw*.

Shaper Size: The size of a shaper is the maximum length of stroke which it can take. Horizontal shapers are most often made with strokes from 175 to 900 mm long, though some smaller and larger sizes are available. The length of the stroke indicates, in addition to the general size of the machine, the size of a cube that can be held and planed in the shaper. Therefore the cross-feed and the maximum height that can be accommodated is approximately equal to the stoke length.

In addition to the stoke length, other particulars like cutting to return stroke ratio, individual motor drive, number and amount of feed etc.

Drive Mechanisms

Shapers are available with either mechanical or hydraulic drive mechanisms. Figures show diagrams of both shaper drive mechanisms.

Mechanical Drive or Crank and Slotted Link Mechanism

The less expensive shaper, the one most often purchased, uses a mechanical drive. Most shaping machine uses Crank and Slotted link mechanism, although sometimes Whitworth quick return mechanism is also used.

Table: The table is moved left and right, usually by hand, to position the work under the cutter when setting up. Then, either by hand or more often automatically, the table is moved sideways to feed the work under the cutter at the end or beginning of each stroke.

Saddle: The saddle moves up and down (Y axis), usually manually, to set the rough position of the depth of cut. Final depth can be set by the hand crank on the tool head.

Column: The column supports the ram and the rails for the saddle. The mechanism for moving the ram and table is housed inside the column.

Tool holders: Tool holders are the same as the ones used on an engine lathe, though often larger in size. The cutter is sharpened with rake and clearance angles similar to lathe tools, though the angles are smaller because the work surface is usually flat. These cutters are fastened into the tool holder, just as in the lathe, but in a vertical plane.

Work Holding: Work holding is frequently done in a vise. The vise is specially designed for use in shapers and has long ways which allow the jaws to open up to 14 inches or more; therefore quite large work pieces can be held. The vise may also have a swivel base so that cuts may be made at an angle. Work which, due to size or shape, cannot be held in the vise, is clamped directly to the shaper table in much the same way as parts are secured on milling machine tables.

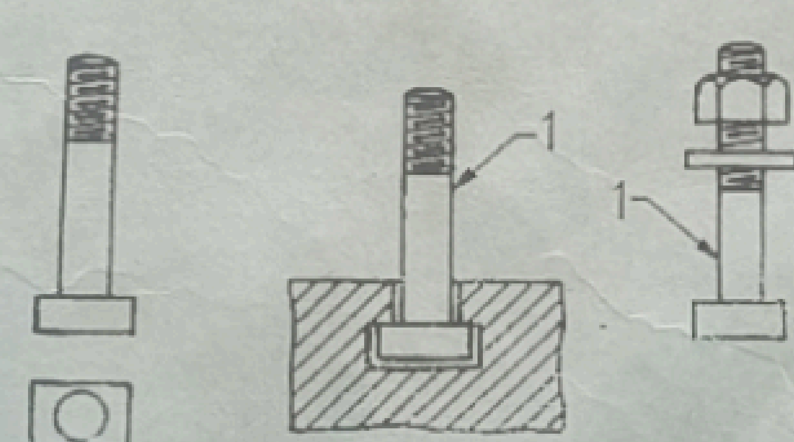


Fig. 5.6 Different views of "T" bolts
1. *T-bolt.*

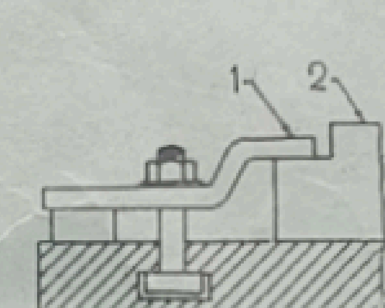


Fig. 5.7 Use of goose neck clamp
1. *Goose neck clamp*, 2. *Work.*

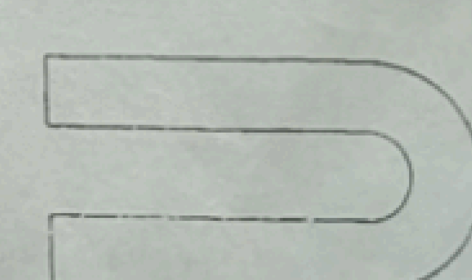


Fig. 5.8 U-clamp.

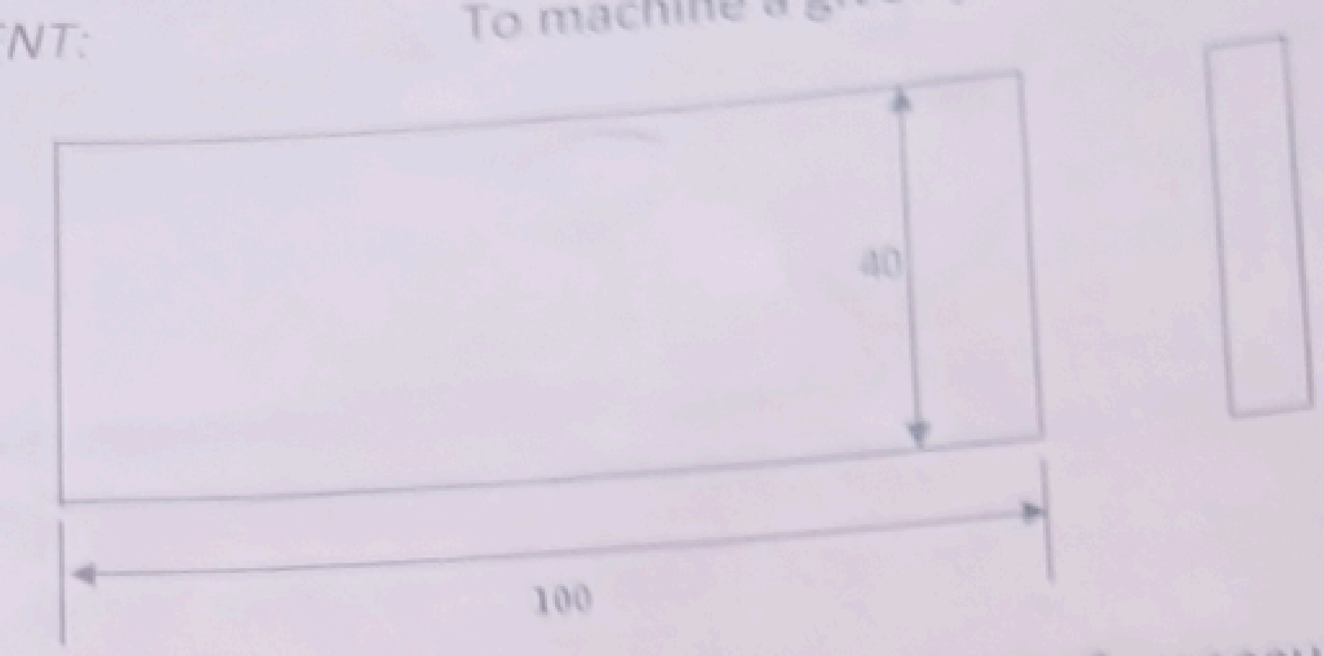
SAFETY PRECAUTIONS FOR MACHINE SHOP

1. NEVER ATTEMPT TO OPERATE ANY MACHINE UNLESS YOU ARE FULLY AWARE OF THE MEANS OF STOPPING THE MACHINE IMMEDIATELY, SHOULD ANY EMERGENCY ARISE.
2. NEVER ENGAGE ANY OPERATING LEVER OR CONTROL UNLESS YOU KNOW IN ADVANCE, ITS DIRECTION OR WHAT ITS EFFECT WILL BE.
3. NEVER IDLY PLAY WITH CONTROL HANDLES ETC. ON ANY MACHINE WHETHER IT IS RUNNING OR STANDING.
4. UNDER NO CIRCUMSTANCES DISTRACT THE ATTENTION OF ANY OTHER OPERATOR WHILST HIS MACHINE IS UNDER POWER.
5. NEVER LEAN AGAINST OR REST THE HANDS ON ANY MOVING PART, EVEN THOUGH THE MOVEMENT BE SLOW, AS YOU MAY EASILY BE TRAPPED UNEXPECTEDLY AT SOME POINT OF ITS TRAVERSE.
6. NEVER OIL OR MAKE ANY ADJUSTMENTS TO A MACHINE, WHICH IS IN MOTION.
7. NEVER START A MACHINE OR ENGAGE A FEED UNTIL YOU ARE CERTAIN THAT THE WORK IS ADEQUATELY SECURED.
8. NEVER LEAVE WRENCHES OR KEYS IN POSITION WHEN SETTING UP OR REMOVING WORK.
9. NEVER CHANGE A BELT POSITION WHILST IT IS RUNNING.
10. ALWAYS REMOVE CHIPS WITH A BRUSH OR OTHER SUITABLE MEANS AND NEVER WITH HANDS OR FINGERS. NEVER ATTEMPT TO BLOW THEM AWAY. YOU MAY EASILY BLOW THEM INTO THE WORKING PARTS OF MACHINE AND FAR MORE EASILY AND LIKELY RECEIVE SOME IN THE EYES.
11. WHEN YOU LEAVE THE MACHINE ALWAYS TURN OFF THE POWER.
12. BEFORE PUTTING ON THE POWER, BE SURE THAT THE WORK IS SECURELY FASTENED.
13. WHILE THE MACHINE IS RUNNING NEVER THROUGH THE BACK GEAR.
14. DO NOT WEAR LOOSE GARMENTS WHILE WORKING ON A MACHINE.

ASSIGNMENT II : SHAPER

OBJECTIVE OF EXPERIMENT: *To study a shaper machine and machine a given job as per given drawing.*

JOB ASSIGNMENT: To machine a given job as per drawing.



TOOLS & INSTRUMENTS: Mention various tools & measuring instruments used in making the assigned job.

PROCESS SHEET: Give details of the process followed.

Process	Description	Tools & instrument

PRECAUTIONS: Mention the necessary safety precautions taken.

QUESTIONS:

1. What is the purpose of a shaping machine?
2. Why the ram of a shaping machine travels at a faster speed during the return stroke?
3. How will you specify the size of a shaping machine?
4. Make a block diagram of shaper machine and explain its principal parts?
5. Explain how an angular cut can be taken in a shaper machine.
6. Explain the functions of clapper block used in a shaper machine.
7. Explain how the stroke length and position of the ram of a shaper machine can be changed.